

to connect local MySQL on windows with HDFS on vm ware machine

Windows 10 host

JDBC

CentOS 6 VM
Sqoop 1.4.7
Hadoop 2.7.3

HDFS

Step 1: Find Windows Host IP : 192.168.1.4

Administrator: Command Prompt

```
C:\WINDOWS\system32>ipconfig
```

Windows IP Configuration

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::77cd:c05f:9d59:d6f0%19
IPv4 Address. : 192.168.246.1
Subnet Mask : 255.255.255.0
Default Gateway :

Ethernet adapter VMware Network Adapter VMnet8:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::e902:9ed:b00:6c47%8
IPv4 Address. : 192.168.226.1
Subnet Mask : 255.255.255.0
Default Gateway :

Wireless LAN adapter Wi-Fi:

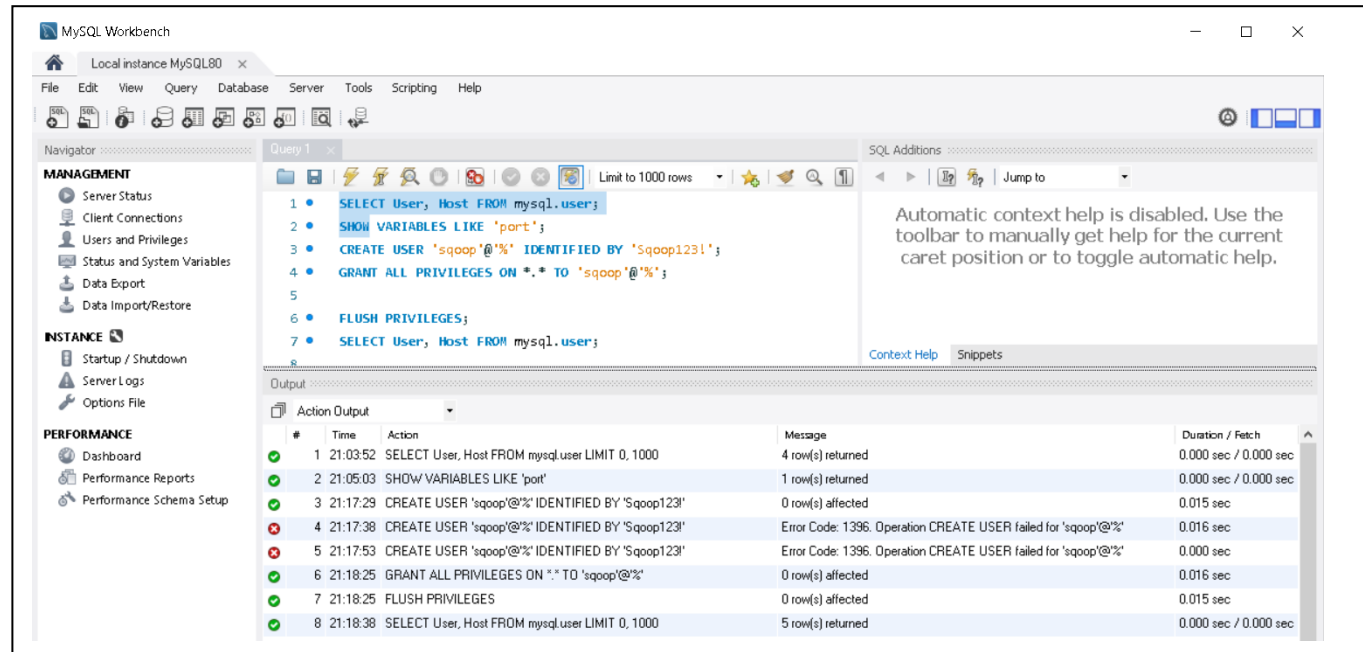
Connection-specific DNS Suffix . :
IPv4 Address. : 192.168.1.4
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :

```
C:\WINDOWS\system32>
```

2- Create a MySQL Database User for Sqoop



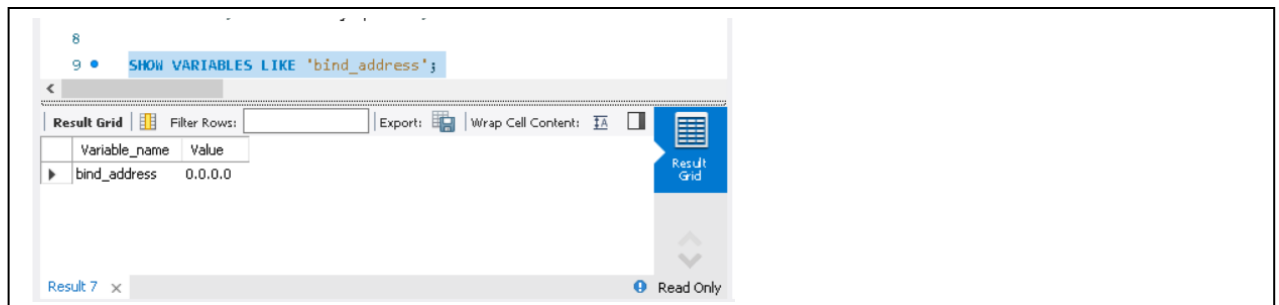
The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' and 'INSTANCE' sections. The main window displays a SQL query with the following commands:

```
1 SELECT User, Host FROM mysql.user;
2 SHOW VARIABLES LIKE 'port';
3 CREATE USER 'sqoop'@'%' IDENTIFIED BY 'Sqoop123!';
4 GRANT ALL PRIVILEGES ON *.* TO 'sqoop'@'%';
5
6 FLUSH PRIVILEGES;
7 SELECT User, Host FROM mysql.user;
```

The 'Output' tab at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	21:03:52	SELECT User, Host FROM mysql.user LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
2	21:05:03	SHOW VARIABLES LIKE 'port'	1 row(s) returned	0.000 sec / 0.000 sec
3	21:17:29	CREATE USER 'sqoop'@'%' IDENTIFIED BY 'Sqoop123!'	0 row(s) affected	0.015 sec
4	21:17:38	CREATE USER 'sqoop'@'%' IDENTIFIED BY 'Sqoop123!'	Error Code: 1396. Operation CREATE USER failed for 'sqoop'@'%'	0.016 sec
5	21:17:53	CREATE USER 'sqoop'@'%' IDENTIFIED BY 'Sqoop123!'	Error Code: 1396. Operation CREATE USER failed for 'sqoop'@'%'	0.000 sec
6	21:18:25	GRANT ALL PRIVILEGES ON *.* TO 'sqoop'@'%'	0 row(s) affected	0.016 sec
7	21:18:25	FLUSH PRIVILEGES	0 row(s) affected	0.015 sec
8	21:18:38	SELECT User, Host FROM mysql.user LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

3: Verify MySQL Listens on Network



The screenshot shows the MySQL Workbench interface with the query 'SHOW VARIABLES LIKE 'bind_address';' executed. The 'Result Grid' tab is active, displaying the following result:

Variable_name	Value
bind_address	0.0.0.0

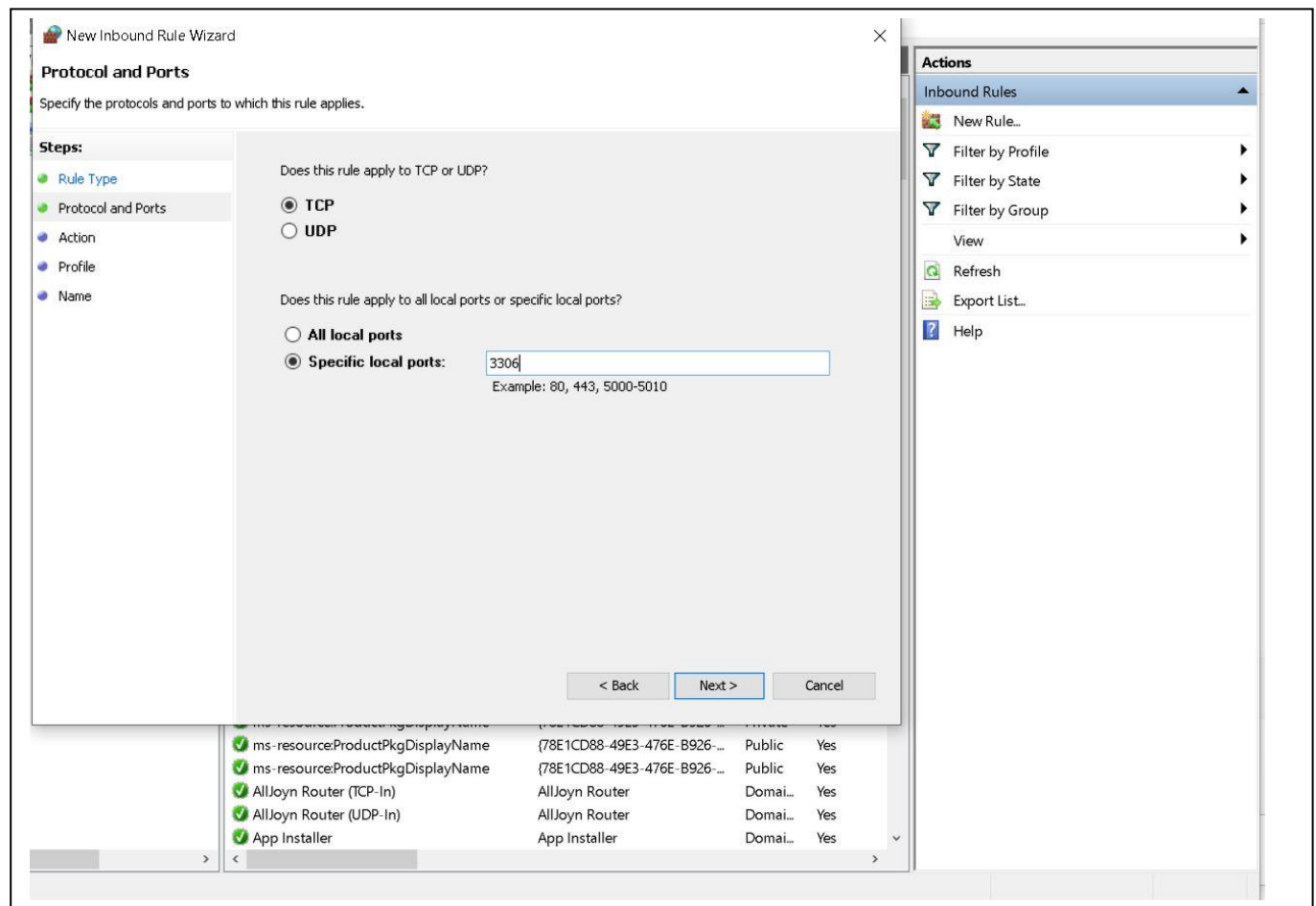
4- Check MySQL Config and add bind-address=0.0.0.0 to "C:\ProgramData\MySQL\MySQL Server 8.0\my.ini" as it was not configured



The screenshot shows a Notepad window titled 'my - Notepad'. The content of the file is as follows:

```
# file.
#
[mysqld]
bind-address=0.0.0.0
```

5- Open Windows Defender Firewall



6-Test Connectivity From CentOS

I used

```
timeout 5 bash -c '</dev/tcp/192.168.1.4/3306' && echo "OPEN" || echo "CLOSED"
```

as telnet WINDOWS_IP 3306 did not work (command not found)

7- I downloaded MySQL JDBC Driver as "sqoop-1.4.7.bin__hadoop-2.6.0" did not include MySQL jar file

MySQL version is 8.0.46 which is an old version so I downloaded "mysql-connector-java-5.1.49.jar" that is more suitable with Sqoop 1.4.7 compatibility

from : "<https://downloads.mysql.com/archives/c-j/>"

Then extracted the downloaded mysql-connector-java-5.1.49 zip file
copy the “mysql-connector-java-5.1.49.jar” to centos directory
“/home/bigdata/sqoop-1.4.7.bin__hadoop-2.6.0/lib”

8- Test MySQL Connection

using command :

```
=> sqoop list-databases --connect jdbc:mysql://192.168.1.4:3306 --  
username sqoop --password 123456!
```

But kept get error (**sqoop: command not found**)

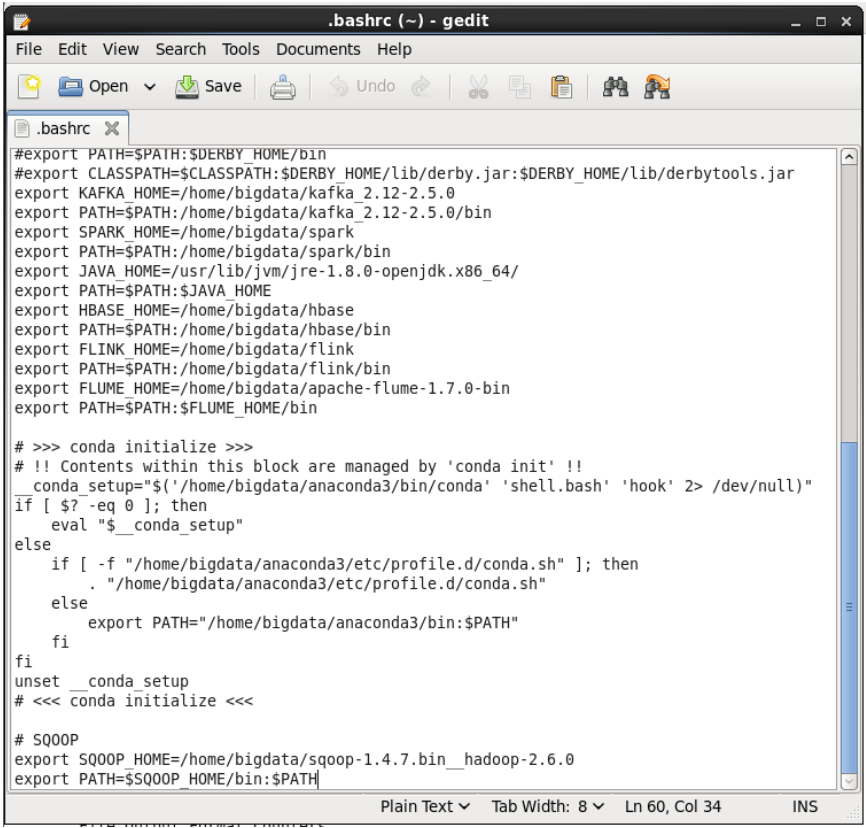
because Sqoop is not in PATH even when running the terminal from
“/home/bigdata/sqoop-1.4.7.bin__hadoop-2.6.0” directory

to put sqoop in path :

```
export SQOOP_HOME=/home/bigdata/sqoop-1.4.7.bin__hadoop-2.6.0
```

```
export PATH=$SQOOP_HOME/bin:$PATH
```

then Edited bash profile for permanent change and sourced it

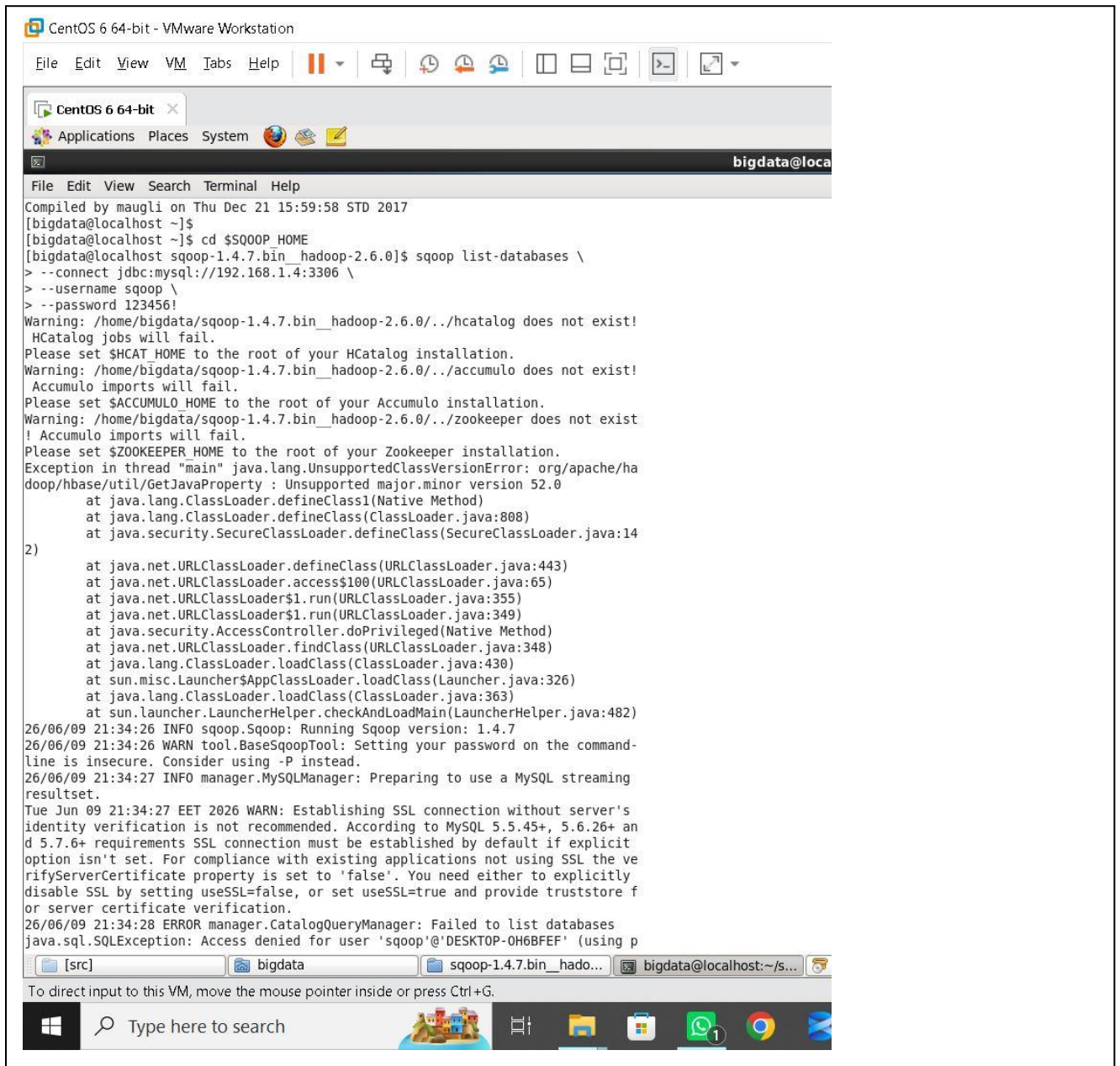


```
.bashrc (~) - gedit
File Edit View Search Tools Documents Help
Open Save Undo Redo Cut Copy Paste
.bashrc
#export PATH=$PATH:$DERBY_HOME/bin
#export CLASSPATH=$CLASSPATH:$DERBY_HOME/lib/derby.jar:$DERBY_HOME/lib/derbytools.jar
export KAFKA_HOME=/home/bigdata/kafka_2.12-2.5.0
export PATH=$PATH:/home/bigdata/kafka_2.12-2.5.0/bin
export SPARK_HOME=/home/bigdata/spark
export PATH=$PATH:/home/bigdata/spark/bin
export JAVA_HOME=/usr/lib/jvm/jre-1.8.0-openjdk.x86_64/
export PATH=$PATH:$JAVA_HOME
export HBASE_HOME=/home/bigdata/hbase
export PATH=$PATH:/home/bigdata/hbase/bin
export FLINK_HOME=/home/bigdata/flink
export PATH=$PATH:/home/bigdata/flink/bin
export FLUME_HOME=/home/bigdata/apache-flume-1.7.0-bin
export PATH=$PATH:$FLUME_HOME/bin

# >>> conda initialize >>>
# !! Contents within this block are managed by 'conda init' !!
__conda_setup="$(/home/bigdata/anaconda3/bin/conda 'shell.bash' 'hook' 2> /dev/null)"
if [ $? -eq 0 ]; then
    eval "$__conda_setup"
else
    if [ -f "/home/bigdata/anaconda3/etc/profile.d/conda.sh" ]; then
        . "/home/bigdata/anaconda3/etc/profile.d/conda.sh"
    else
        export PATH="/home/bigdata/anaconda3/bin:$PATH"
    fi
fi
unset __conda_setup
# <<< conda initialize <<<

# SQOOP
export SQOOP_HOME=/home/bigdata/sqoop-1.4.7.bin__hadoop-2.6.0
export PATH=$SQOOP_HOME/bin:$PATH
```

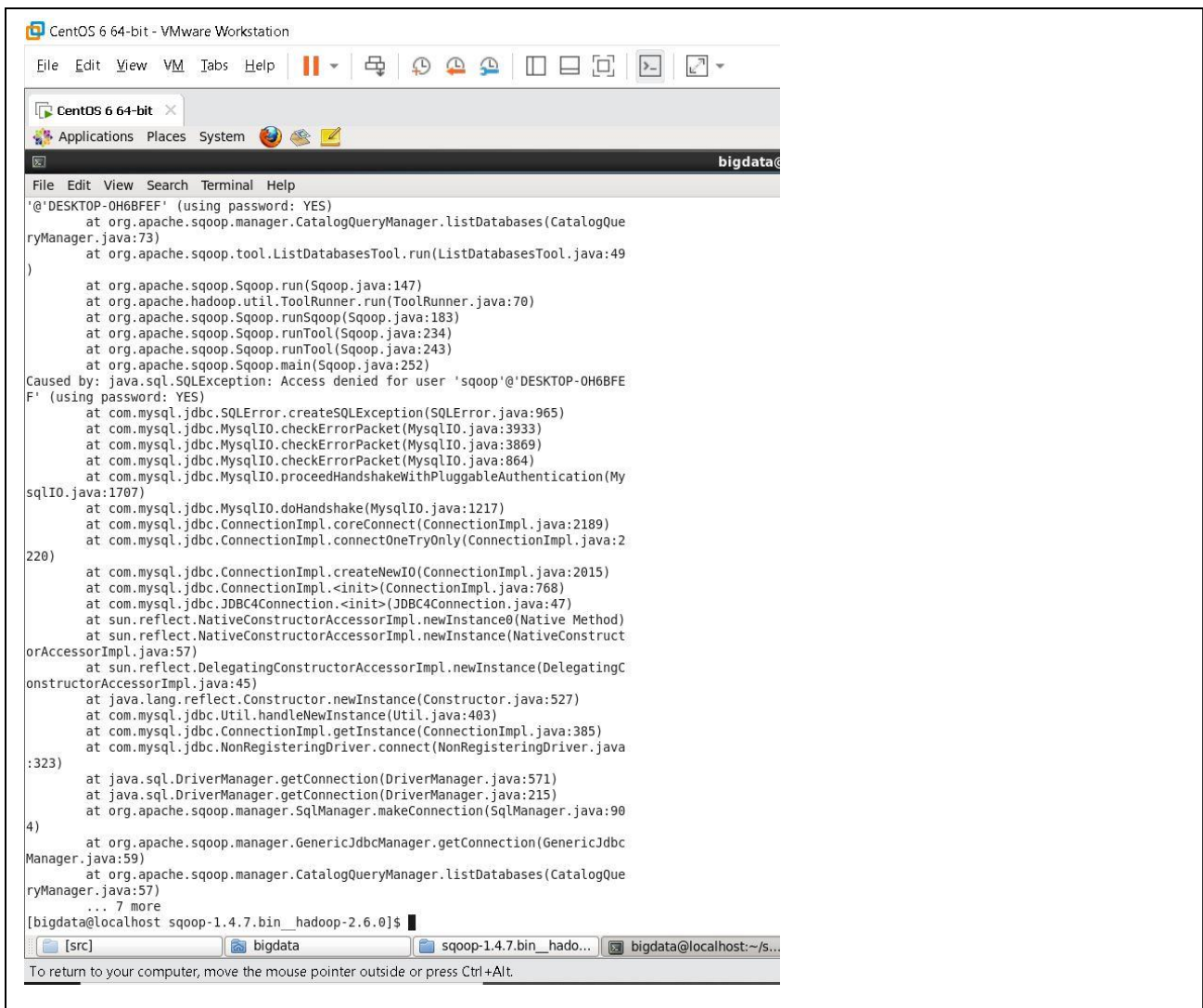
back to testing connection: **it is connected**



The screenshot shows a VMware Workstation window titled "CentOS 6 64-bit - VMware Workstation". Inside the VM, a terminal window is open with the prompt "bigdata@localhost". The terminal output shows the following commands and messages:

```
Compiled by maugli on Thu Dec 21 15:59:58 STD 2017
[bigdata@localhost ~]$
[bigdata@localhost ~]$ cd $SQOOP_HOME
[bigdata@localhost sqoop-1.4.7.bin_hadoop-2.6.0]$ sqoop list-databases \
> --connect jdbc:mysql://192.168.1.4:3306 \
> --username sqoop \
> --password 123456!
Warning: /home/bigdata/sqoop-1.4.7.bin_hadoop-2.6.0/./hcatalog does not exist!
HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /home/bigdata/sqoop-1.4.7.bin_hadoop-2.6.0/./accumulo does not exist!
Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /home/bigdata/sqoop-1.4.7.bin_hadoop-2.6.0/./zookeeper does not exist!
Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
Exception in thread "main" java.lang.UnsupportedClassVersionError: org/apache/ha
doop/hbase/util/GetJavaProperty : Unsupported major.minor version 52.0
    at java.lang.ClassLoader.defineClass1(Native Method)
    at java.lang.ClassLoader.defineClass(ClassLoader.java:808)
    at java.security.SecureClassLoader.defineClass(SecureClassLoader.java:14
2)
    at java.net.URLClassLoader.defineClass(URLClassLoader.java:443)
    at java.net.URLClassLoader.access$100(URLClassLoader.java:65)
    at java.net.URLClassLoader$1.run(URLClassLoader.java:355)
    at java.net.URLClassLoader$1.run(URLClassLoader.java:349)
    at java.security.AccessController.doPrivileged(Native Method)
    at java.net.URLClassLoader.findClass(URLClassLoader.java:348)
    at java.lang.ClassLoader.loadClass(ClassLoader.java:430)
    at sun.misc.Launcher$AppClassLoader.loadClass(Launcher.java:326)
    at java.lang.ClassLoader.loadClass(ClassLoader.java:363)
    at sun.launcher.LauncherHelper.checkAndLoadMain(LauncherHelper.java:482)
26/06/09 21:34:26 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
26/06/09 21:34:26 WARN tool.BaseSqoopTool: Setting your password on the command-
line is insecure. Consider using -P instead.
26/06/09 21:34:27 INFO manager.MySQLManager: Preparing to use a MySQL streaming
resultset.
Tue Jun 09 21:34:27 EET 2026 WARN: Establishing SSL connection without server's
identity verification is not recommended. According to MySQL 5.5.45+, 5.6.26+ an
d 5.7.6+ requirements SSL connection must be established by default if explicit
option isn't set. For compliance with existing applications not using SSL the ve
rifyServerCertificate property is set to 'false'. You need either to explicitly
disable SSL by setting useSSL=false, or set useSSL=true and provide truststore f
or server certificate verification.
26/06/09 21:34:28 ERROR manager.CatalogQueryManager: Failed to list databases
java.sql.SQLException: Access denied for user 'sqoop'@'DESKTOP-0H6BFEF' (using p
```

The terminal window has a menu bar with "File", "Edit", "View", "Search", "Terminal", and "Help". The bottom of the window shows a taskbar with icons for Windows, search, and several applications. The status bar at the bottom says "To direct input to this VM, move the mouse pointer inside or press Ctrl+G."



9- try listing DB from MySQL in HDFS

=> sqoop list-databases --connect jdbc:mysql://192.168.1.4:3306 --
username sqoop --password 123456!

I got a **Hadoop/YARN misconfiguration problem !**

while running the last command I got

Connecting to ResourceManager at /0.0.0.0:8032

Retrying connect to server: 0.0.0.0:8032

as Hadoop client configuration is pointing ResourceManager to **0.0.0.0 (INVALID address)** and Sqoop (map-reduce) keeps trying forever

so , I edited “\$HADOOP_HOME/etc/hadoop/yarn-site.xml” file and added configuration

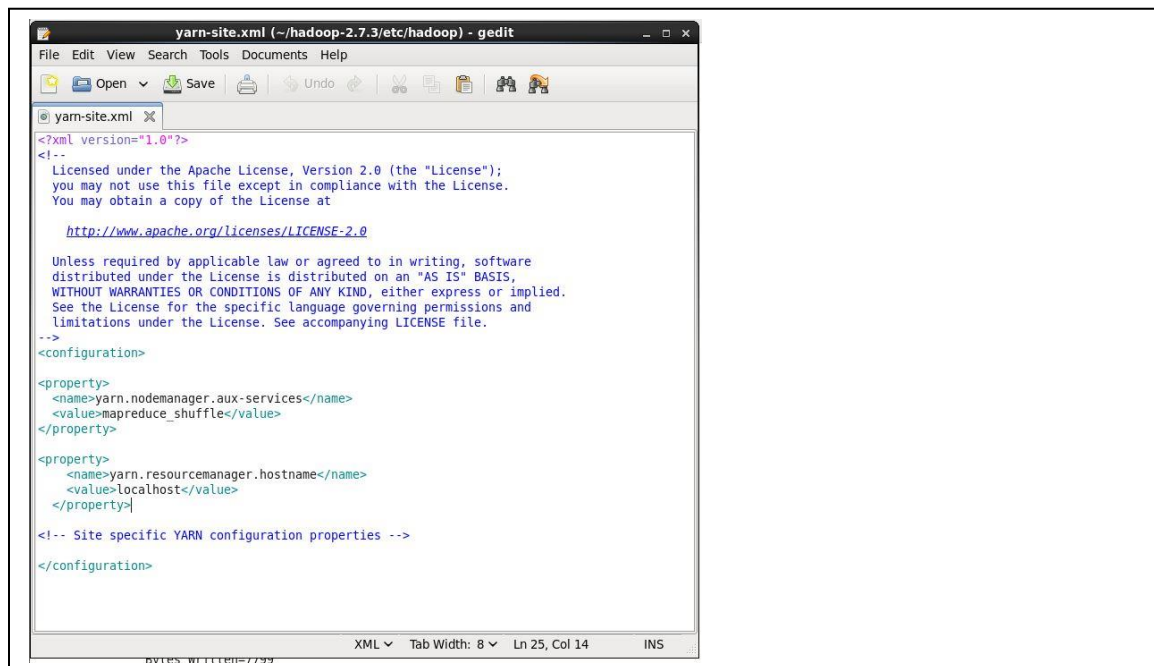
```
<property>
```

```
<name>yarn.resourcemanager.hostname</name>
```

```
<value>localhost</value>
```

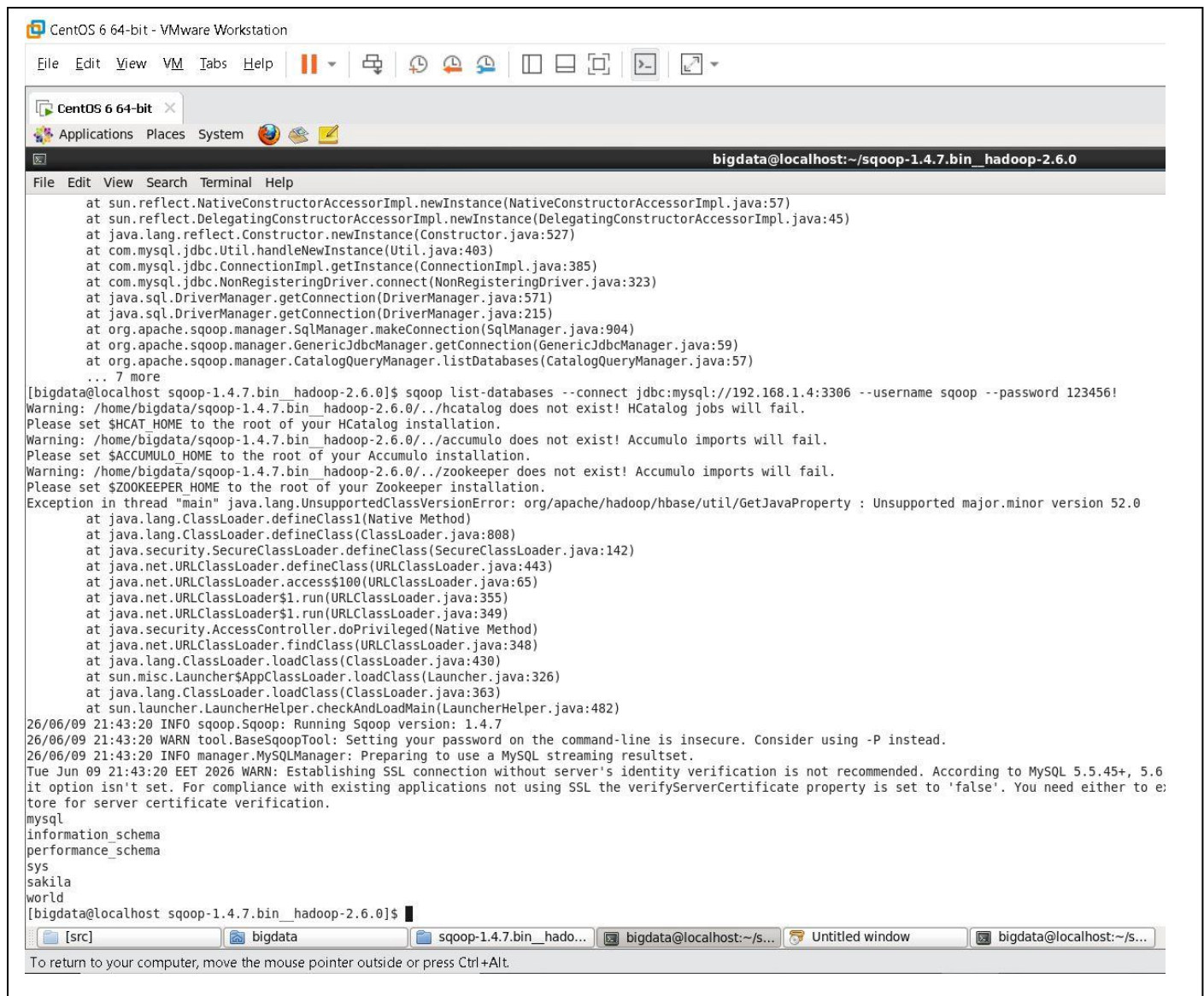
```
</property>
```

as it was not involved in the XML file



Then restarting HADOOP (I actually restarted the all machine and start HDFS and yarn again)

back sqoop listing MySQL databases: **it worked**



The screenshot shows a CentOS 6 64-bit VM window titled "CentOS 6 64-bit - VMware Workstation". The terminal window is titled "bigdata@localhost:~/sqoop-1.4.7.bin_hadoop-2.6.0". The terminal output shows the execution of the command `sqoop list-databases --connect jdbc:mysql://192.168.1.4:3306 --username sqoop --password 123456!`. The output includes several warnings about HCatalog and Accumulo, followed by an exception: `Exception in thread "main" java.lang.UnsupportedClassVersionError: org/apache/hadoop/hbase/util/GetJavaProperty : Unsupported major.minor version 52.0`. The exception details show the class loader path and the version mismatch. The terminal also shows the output of the `mysql` command, listing the databases: `information_schema`, `performance_schema`, `sys`, `sakila`, and `world`.

```
at sun.reflect.NativeConstructorAccessorImpl.newInstance(NativeConstructorAccessorImpl.java:57)
at sun.reflect.DelegatingConstructorAccessorImpl.newInstance(DelegatingConstructorAccessorImpl.java:45)
at java.lang.reflect.Constructor.newInstance(Constructor.java:527)
at com.mysql.jdbc.Util.handleNewInstance(Util.java:403)
at com.mysql.jdbc.ConnectionImpl.getInstance(ConnectionImpl.java:385)
at com.mysql.jdbc.NonRegisteringDriver.connect(NonRegisteringDriver.java:323)
at java.sql.DriverManager.getConnection(DriverManager.java:571)
at java.sql.DriverManager.getConnection(DriverManager.java:215)
at org.apache.sqoop.manager.SqlManager.makeConnection(SqlManager.java:904)
at org.apache.sqoop.manager.GenericJdbcManager.getConnection(GenericJdbcManager.java:59)
at org.apache.sqoop.manager.CatalogQueryManager.listDatabases(CatalogQueryManager.java:57)
... 7 more
[bigdata@localhost sqoop-1.4.7.bin_hadoop-2.6.0]$ sqoop list-databases --connect jdbc:mysql://192.168.1.4:3306 --username sqoop --password 123456!
Warning: /home/bigdata/sqoop-1.4.7.bin_hadoop-2.6.0/./hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /home/bigdata/sqoop-1.4.7.bin_hadoop-2.6.0/./accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /home/bigdata/sqoop-1.4.7.bin_hadoop-2.6.0/./zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
Exception in thread "main" java.lang.UnsupportedClassVersionError: org/apache/hadoop/hbase/util/GetJavaProperty : Unsupported major.minor version 52.0
at java.lang.ClassLoader.defineClass1(Native Method)
at java.lang.ClassLoader.defineClass(ClassLoader.java:808)
at java.security.SecureClassLoader.defineClass(SecureClassLoader.java:142)
at java.net.URLClassLoader.defineClass(URLClassLoader.java:443)
at java.net.URLClassLoader.access$100(URLClassLoader.java:65)
at java.net.URLClassLoader$1.run(URLClassLoader.java:355)
at java.net.URLClassLoader$1.run(URLClassLoader.java:349)
at java.security.AccessController.doPrivileged(Native Method)
at java.net.URLClassLoader.findClass(URLClassLoader.java:348)
at java.lang.ClassLoader.loadClass(ClassLoader.java:430)
at sun.misc.Launcher$AppClassLoader.loadClass(Launcher.java:326)
at java.lang.ClassLoader.loadClass(ClassLoader.java:363)
at sun.launcher.LauncherHelper.checkAndLoadMain(LauncherHelper.java:482)
26/06/09 21:43:20 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
26/06/09 21:43:20 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
26/06/09 21:43:20 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
Tue Jun 09 21:43:20 EET 2026 WARN: Establishing SSL connection without server's identity verification is not recommended. According to MySQL 5.5.45+, 5.6
it option isn't set. For compliance with existing applications not using SSL the verifyServerCertificate property is set to 'false'. You need either to e
tore for server certificate verification.
mysql
information_schema
performance_schema
sys
sakila
world
[bigdata@localhost sqoop-1.4.7.bin_hadoop-2.6.0]$
```

10- listing table from MySQL on windows 10 on HDFS on centos 6

sqoop import --connect jdbc:mysql://192.168.1.4:3306/sakila --username sqoop -
password 123456! --table actor --target-dir /db_from_windows --driver
com.mysql.jdbc.Driver --mapreduce-job-name "mysql_to_hdfs_actor"

It worked!

The screenshot shows a VMware Workstation interface with a single virtual machine named 'CentOS 6 64-bit'. The terminal window is open, displaying the output of the 'hadoop job -list' command. The output shows a completed job with various counters and metrics.

```

mode : false
26/06/09 22:25:14 INFO mapreduce.Job: map 0% reduce 0%
26/06/09 22:25:50 INFO mapreduce.Job: map 25% reduce 0%
26/06/09 22:25:51 INFO mapreduce.Job: map 50% reduce 0%
26/06/09 22:25:52 INFO mapreduce.Job: map 100% reduce 0%
26/06/09 22:25:53 INFO mapreduce.Job: Job job_1781036588706_0001 completed successfully
26/06/09 22:25:54 INFO mapreduce.Job: Counters: 31
  File System Counters
    FILE: Number of bytes read=0
    FILE: Number of bytes written=555500
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=437
    HDFS: Number of bytes written=7799
    HDFS: Number of read operations=16
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=8
  Job Counters
    Killed map tasks=1
    Launched map tasks=4
    Other local map tasks=4
    Total time spent by all maps in occupied slots (ms)=129600
    Total time spent by all reduces in occupied slots (ms)=0
    Total time spent by all map tasks (ms)=129600
    Total vcore-milliseconds taken by all map tasks=129600
    Total megabyte-milliseconds taken by all map tasks=132710400
  Map-Reduce Framework
    Map input records=200
    Map output records=200
    Input split bytes=437
    Spilled Records=0
    Failed Shuffles=0
    Merged Map outputs=0
    GC time elapsed (ms)=9094
    CPU time spent (ms)=8250
    Physical memory (bytes) snapshot=746364928
    Virtual memory (bytes) snapshot=4077711360
    Total committed heap usage (bytes)=517996544
  File Input Format Counters
    Bytes Read=0
  File Output Format Counters
    Bytes Written=7799
26/06/09 22:25:54 INFO mapreduce.ImportJobBase: Transferred 7.6162 KB in 74.4953 seconds (104.6912 bytes/sec)
26/06/09 22:25:54 INFO mapreduce.ImportJobBase: Retrieved 200 records.
(base) [bigdata@localhost sqoop-1.4.7.bin__hadoop-2.6.0]$
  
```

Hadoop

Overview

Datanodes

Snapshot

Startup Progress

Utilities

Browse Directory

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
-rw-r--r--	bigdata	supergroup	0 B	Tue 09 Jun 2026 10:25:51 PM EET	1	128 MB	_SUCCESS
-rw-r--r--	bigdata	supergroup	1.87 KB	Tue 09 Jun 2026 10:25:48 PM EET	1	128 MB	part-m-00000
-rw-r--r--	bigdata	supergroup	1.88 KB	Tue 09 Jun 2026 10:25:50 PM EET	1	128 MB	part-m-00001
-rw-r--r--	bigdata	supergroup	1.93 KB	Tue 09 Jun 2026 10:25:50 PM EET	1	128 MB	part-m-00002
-rw-r--r--	bigdata	supergroup	1.93 KB	Tue 09 Jun 2026 10:25:51 PM EET	1	128 MB	part-m-00003

Hadoop, 2016.